Outline:

1. *Models used*
2. *Bayesian estimation results*
3. *Output gap results*
4. *Ideas for future research*

1. The models and techniques used

There are several ways to estimate/ calculate unobserved variables such as the output gap and potential output growth. I have used 3 different techniques to calculate the overall and non-agricultural output gap in Rwanda:

- HP filter:
  - separates output into a trend and a cyclical component
  - lambda determines the smoothness of the trend. The higher lambda, the more the trend resembles a linear time trend.
- Band-Pass (BP) filter:
  - Separates output into components based on frequency in which fluctuations occur
- Kalman filter:
  - Uses a dynamic model to estimate the unknown variables
  - The models have 5 inputs
    - Observable variables
    - Unobservable variables (estimated using the Kalman filter)
    - Set parameters
    - Estimated parameters (estimated using Bayesian estimation)
    - Estimated standard error distribution (estimated using Bayesian estimation)
  - I've used 3 different dynamic models to estimate output gap as summarized in the table below
    - One with only output as the observable variable
    - Two with both output and inflation as inputs
      - Normal Philips-curve
      - Lagged Philips-curve

<u>Data and Results</u>

- Quarterly data on output and core inflation from 2009Q2-2019Q4 is used
  - Data is seasonally adjusted
  - Due to covid, I have decided to omit 2020 data
- Results are presented for 2010Q1-2019Q4
- All methods are run twice, once for overall output and once using just non-agricultural output
- For each method smoothed and unsmoothed results are calculated
  - Smooth output gap: utilizes all historical data to calculate gap at time x, also data after x
  - Unsmooth output gap: only utilizes data upto time x to calculate output gap at x

- In last period of data availability: smooth output gap = unsmooth output gap

Table 1: Models used

Model	Equations	Definitions	Set par	Est. par
HP-filter			Lambda = 1600	
BP-filter	Fluctuation horizons: Output gap: 6-32 quarters Natural output: 32-infinite quarters			
Kalman filter, one variable	(1) potential output definition: $y_t = \acute{y}_t + \hat{y}_t$ (2) stoc process for potential output: $\acute{y}_t = \acute{y}_{t-1} + SSG_t + \varepsilon_t$ (3) stoc process for SS potential growth rate: $SSG_t =$ ¿ $SSG_{t-1} + (1-\rho) * \acute{SSG} + \eta_t$ (4) stoc process for output gap: $\hat{y}_t = \alpha * \hat{y}_{t-1} + \xi_t$	<u>Variables:</u> y_t = log real output $\acute{y}_t$ = potential output $\hat{y}_t$ = output gap SSG = Long run potential growth rate <u>Parameters:</u> $\acute{SSG}$ = SS potential growth rate ρ = potential growth AR(1) process parameter α = output gap AR(1) parameter ε_t = shock to potential output η_t = shock to LR potential growth rate ξ_t = shock to output gap	$\acute{SSG}$ = avg. Q growth rate of output (1.58 for all output and 1.72 for NA output)	ρ α ε_t η_t ξ_t
Kalman filter, two variables (with Philips-curve)	Equations (1)-(4) + V1: (5a) current gap PC: $\pi_t = \theta * \pi_{t+1} + (1-\theta) * \pi_{t-1} + \beta * \hat{y}_t + v_t$ V2: (5b) lagged gap PC: $\pi_t = \theta * \pi_{t+1} + (1-\theta) * \pi_{t-1} + \beta * \hat{y}_{t-1} + v_t$	<u>Variables:</u> π_t = annualised core inflation π_{t+1} = rational expectation inflation at t+1 <u>Parameters:</u> θ = % of forward-looking agents β = effect of output gap on inflation v_t = shock to inflation		θ β v_t

2. **Bayesian estimation results**

What is Bayesian estimation:

Bayesian inference techniques normally refer to a method in which unknowns (posteriors) are inferred by minimizing a loss function given a set of prior beliefs/information. When estimating parameters, we use a form of maximum likelihood estimation to find the point of highest probability.

In very simplified terms this works as follows: let's imagine you are in a landscape and you are told find the 'highest point' in this landscape. The problem is that you do not have birds-eye view of the landscape, you can only look at a limited area around your feet. In order to find the highest point, you simply walk in the direction which is upward sloping. A maximum is reached once you reach a point where the area is downward sloping in all direction.

Issues with this method mostly arise from the fact that we don't know whether a local or global maximum is found. This is where the importance of prior beliefs becomes clear. Priors give the starting point in the landscape. And hence, the closer the priors are to the 'true' parameter values, the more likely it is that the posteriors values reflect the true values. In a simple system of equations, priors have little influence on the posterior values. As the system of equations becomes more complicated, the more important priors are in determining posterior values.

Prior Inputs in Bayesian estimation:

Each estimated parameter requires the following priors

- Mean
- Upper and lower boundaries
- Standard deviation

The entire dataset is used for the Bayesian estimation

Results

Table 2 shows the results from the Bayesian estimation for the 8 unknown parameter values.

- First column refers to mean prior value
- Bayesian estimation has been done for 6 different models
    - 3 systems of equations as outlined in table 1
        - KF1: simplest model without Philips curve, only uses output data
        - KF2: uses both output and inflation (V1 in table 1)
        - KF3: uses both output and inflation but in the Philips curve lagged output gap impacts inflation (V2 in table 1)
    - Each system of equations is run two times. One time using total GDP (all) and once only using non-agricultural GDP (NA) as the output variable

Table 2: Bayesian estimation priors and posteriors

parameter	prior	post						Sugg By arend
		KF1 (without PC)		KF2 (with PC)		KF3 (with lagged PC)		
		all	NA	all	NA	all	NA	
ρ = potential growth AR(1) process parameter	0.8	0.73	0.75	0.81	0.82	0.81	0.82	$(1-\rho¿¿y)¿$:0.3–0.62
α = output gap AR(1) parameter	0.6	0.59	0.55	0.56	0.52	0.56	0.51	-
θ = % of forward-looking agents	0.25	-	-	0.35	0.35	0.33	0.33	β_1: 0.175 – 0.5
β = effect of output gap on inflation	0.25	-	-	0.16	0.13	0.21	0.21	β_2: 0.2 – 0.41
ε_t = shock to potential output	0.35	0.35	0.35	0.35	0.35	0.35	0.35	-
η_t = shock to LR potential growth rate	0.35	0.33	0.34	0.33	0.34	0.33	0.34	-
ξ_t = shock to output gap	1.6	1.59	1.61	1.59	1.61	1.59	1.61	-
ν_t = shock to inflation	1.9	-	-	1.91	1.91	1.91	1.91	-

ρ = potential growth AR(1) process parameter

- Indicates speed with which potential growth convergences towards its steady-state value. Our values are rather high, indicating a slow speed of convergence
- Values are higher for the models which include inflation relationships, as output gaps become less dependent on SSG. This allows SSG to be less volatile
- Values are pretty similar between ALL and NA although those for NA tend to be higher. This is unsurprising as agricultural growth tends to be more constant.
- Current values are outside Arend's suggested range. This could be due to the fact that $S\acute{S}G$ is more variable in Arend's model, whereas in my models it is constant.

α = output gap AR(1) parameter

- Indicates 'smoothness'/ volatility of output gap
- Models with PCs have slightly lower values
- Values for NA output are slightly lower

θ = % of forward-looking agents

- Values are pretty stable across methods and output definitions. Values are within Arend's range

β = effect of output gap on core inflation

- For the normal PC, values are very low. It appears that the PC relationship between inflation and output gap is very weak. These values are outside of Arend's range. The fact that Arend's PC includes more variables does not explain this. If anything, one would expect this difference to actually result in a higher value for our beta compared to Arend's

- To make matters worse, beta is LOWER for NA output than all output. I definitely expected the relationship between NA output gap and inflation to be stronger than that between all output and inflation given the negative effect of agricultural shocks on prices
- The weak relationship between inflation and output gap could be due to the fact that a quarter is too short of a time period for prices to react to the output gap. Hence, I ran another model whith lagged output gap in the PC.
    - This does increase beta to 0.21, equal to the figure from NBR (2015) cited by Arend
    - The beta for NA is still not larger than the beta for overall output

Standard errors of shocks

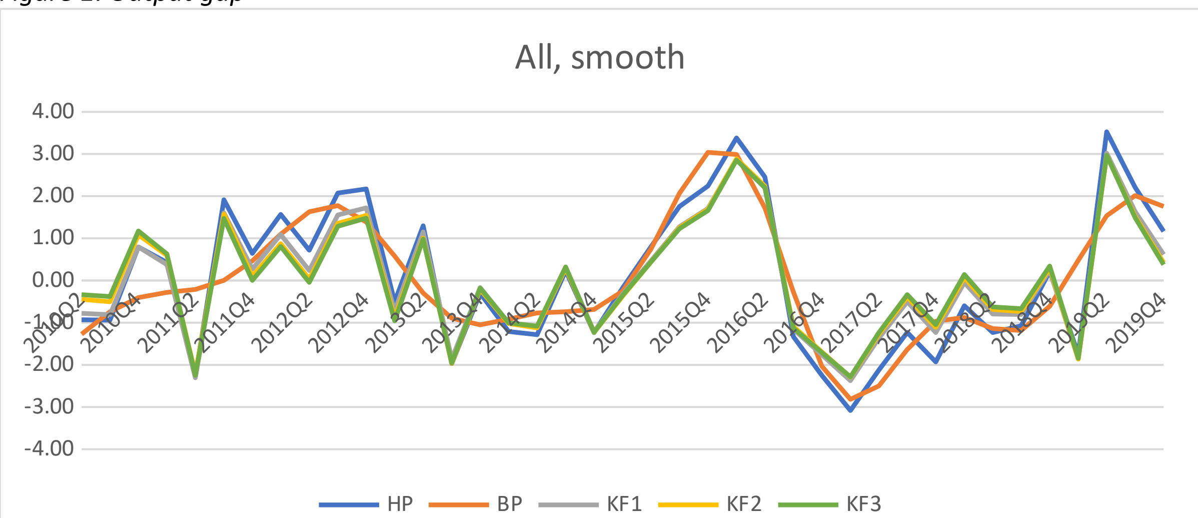
- SE for shocks remain pretty constant throughout models and output type
- SE figures in table 2 will be used in relative terms, not absolute

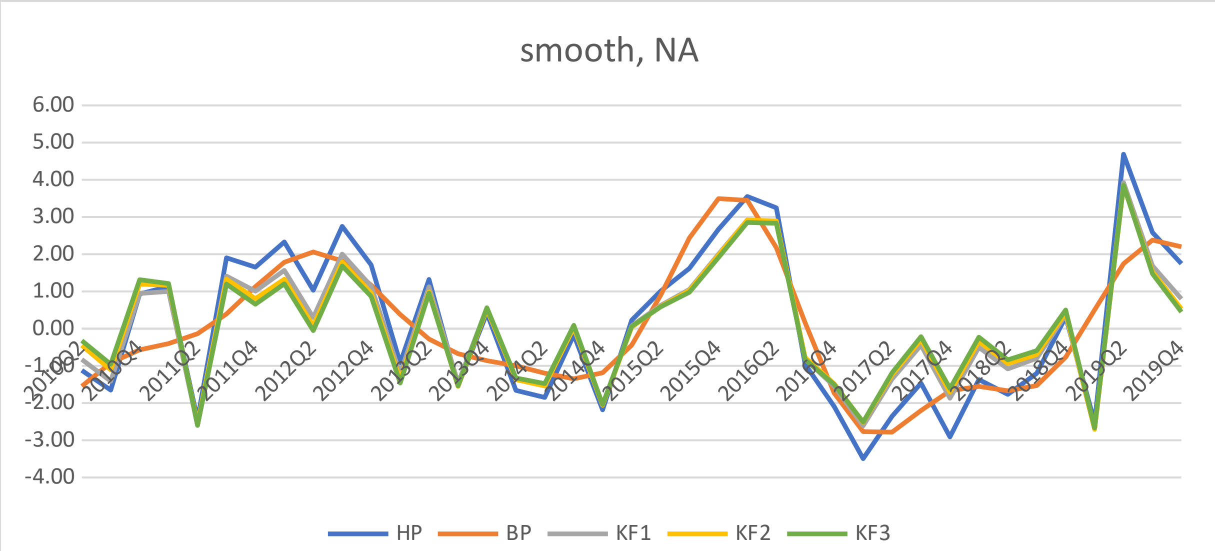
3. Results

A. *Output gap results*

- Figure 1 shows time-series results for the output gap for both all output and NA output, for smoothed and end-point results
    - Output gap results for the three KF models are very similar to each other
        - for all 4 categories
        - For end-point, we see slight difference between KF1 and KF2+KF3 during the 2010-2014 period
    - HP filter
        - For the smooth results, hp filter is pretty close to the KF results. In the instances where they differ, generally the HP filter is more extreme/ further away from zero
        - For the end point results, hp filter and KF results differ more often
    - BP filter is less volatile than other methods by design.
        - The smoothed results appear to be a pretty good approximation of a moving average
        - The end-point results are more volatile than the smoothed. Results also appear to be a bit behind development

Figure 1: Output gap




smooth, NA
6.00
5.00
4.00
3.00
2.00
1.00
0.00
-1.00
-2.00
-3.00
-4.00
2010Q2
2010Q4
2011Q2
2011Q4
2012Q2
2012Q4
2013Q2
2013Q4
2014Q2
2014Q4
2015Q2
2015Q4
2016Q2
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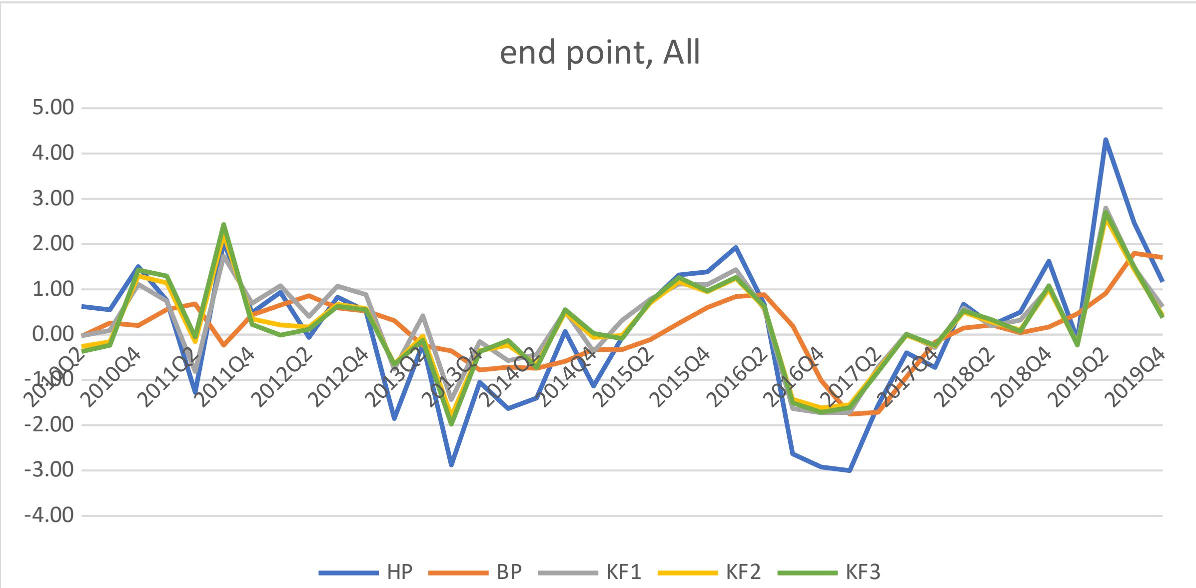
2019Q2
2019Q4
HP
BP
KF1
KF2
KF3


end point, All
5.00
4.00
3.00
2.00
1.00
0.00
-1.00
-2.00
-3.00
-4.00
2010Q2
2010Q4
2011Q2
2011Q4
2012Q2
2012Q4
2013Q2
2013Q4
2014Q2
2014Q4
2015Q2
2015Q4
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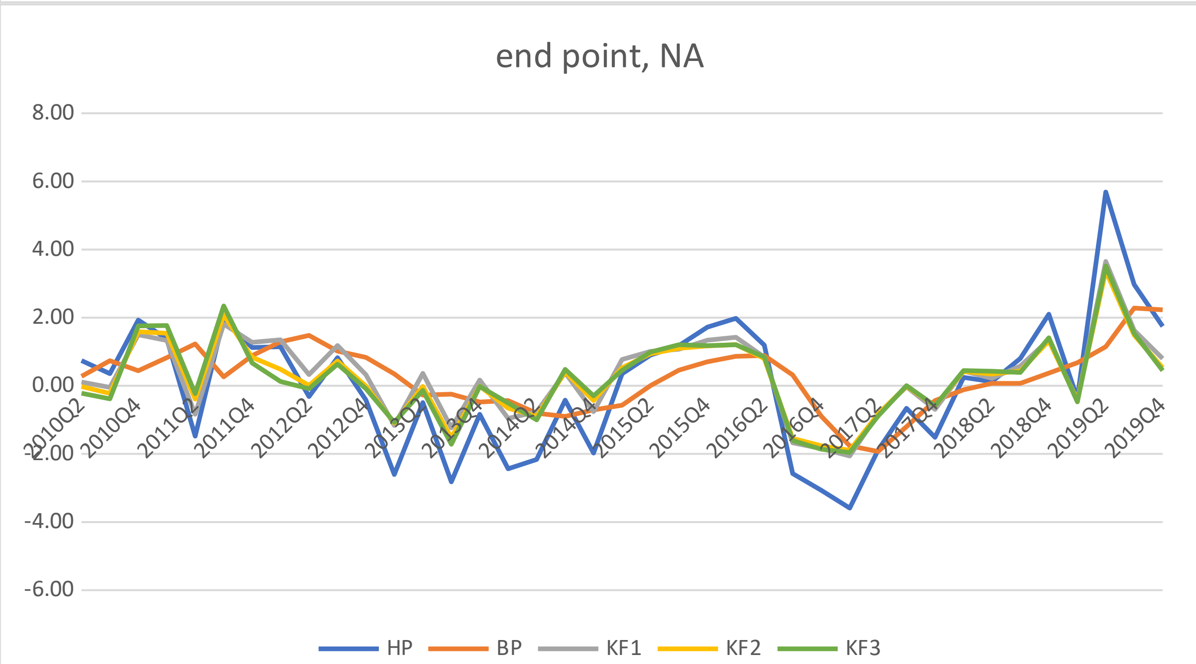
HP
BP
KF1
KF2
KF3


end point, NA
8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
-6.00
2010Q2
2010Q4
2011Q2
2011Q4
2012Q2
2012Q4
2013Q2
2013Q4
2014Q2
2014Q4
2015Q2
2015Q4
2016Q2
2016Q4
2017Q2
2017Q4
2018Q2
2018Q4
2019Q2
2019Q4
HP
BP
KF1
KF2
KF3

- Table 3 and 4 show the difference between the end-point and smoothed results.
    - If there is a large difference between end-point and smooth results, output gap results at the end of the sample period become unreliable.
        - The HP filter is notorious for unreliable end-point results
    - Two methods are used to evaluate the accuracy/reliability of end-point results.
    - The KF models represent a large improvement in terms of end-point accuracy compared to both HP and BP filters.
        - This, in combination with the fact that KF and HP smoothed filter results showcase only small differences, indicates that using a KF filter to calculate output gaps represents an improvement on using the HP filter
    - Within the KF models, including the Philips curve (KF2+KF3) does not improve end-point accuracy. And in most cases actually worsens end-point accuracy.
        - According to literature, the KF2 models should be an improvement on the KF1 model. Our results are therefore a little counterintuitive and could be due to the following reasons:
            - The relationship between inflation and output gap is too weak and therefore including the PC just increases noise
            - The PC requires a forecast for next-period inflation. This forecast is based on the output gap and current inflation. However the current model is bad at predicting next-period inflation because the PC relationship is weak and quarterly inflation is volatile
        - The fact that KF3 does not have higher accuracy than KF1 is more intuitive as the PC curve relationship has no direct influence on the end-point estimate of the output gap. This is due to the fact that a Kalman filter only provides one period ahead forecasts and the end-point KF3 PC curve would require two period ahead forecasts. The end-point estimates between KF3 and KF1 differ because of the different Bayesian estimates of parameter values

Table 3: Difference between smooth and end-point output gap estimate, all output

method	HP	BP	KF1	KF2	KF3
Sum of absolute values	34.4	29.5	23.4	23.9	23.5
Sum of squares	42.3	34.5	21.4	23.7	24.3

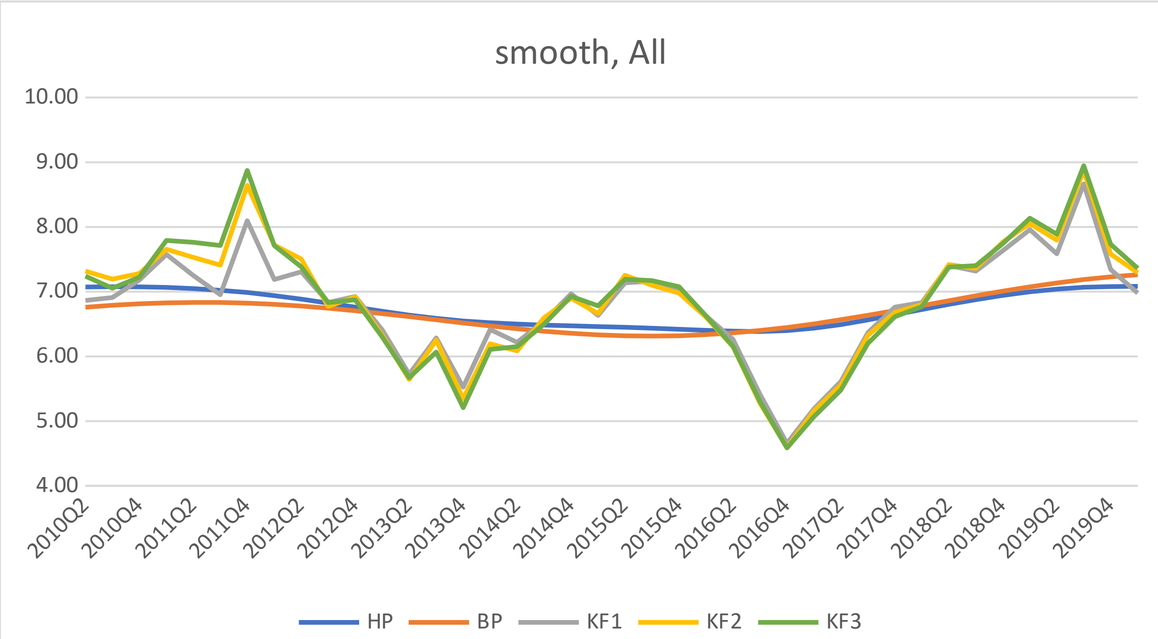
Table 4: Difference between smooth and end-point output gap estimate, non-agricultural output

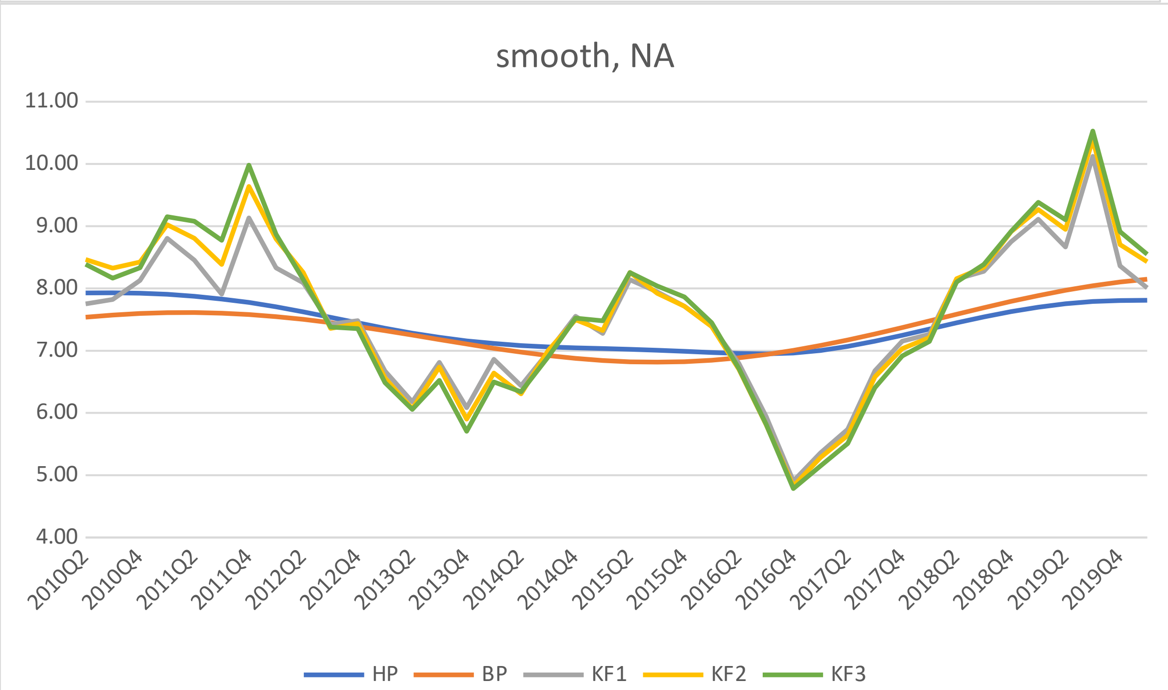
method	HP	BP	KF1	KF2	KF3
Sum of absolute values	44.3	35.7	28.7	29.5	29.4
Sum of squares	68.7	51.3	32.7	35.2	35.5

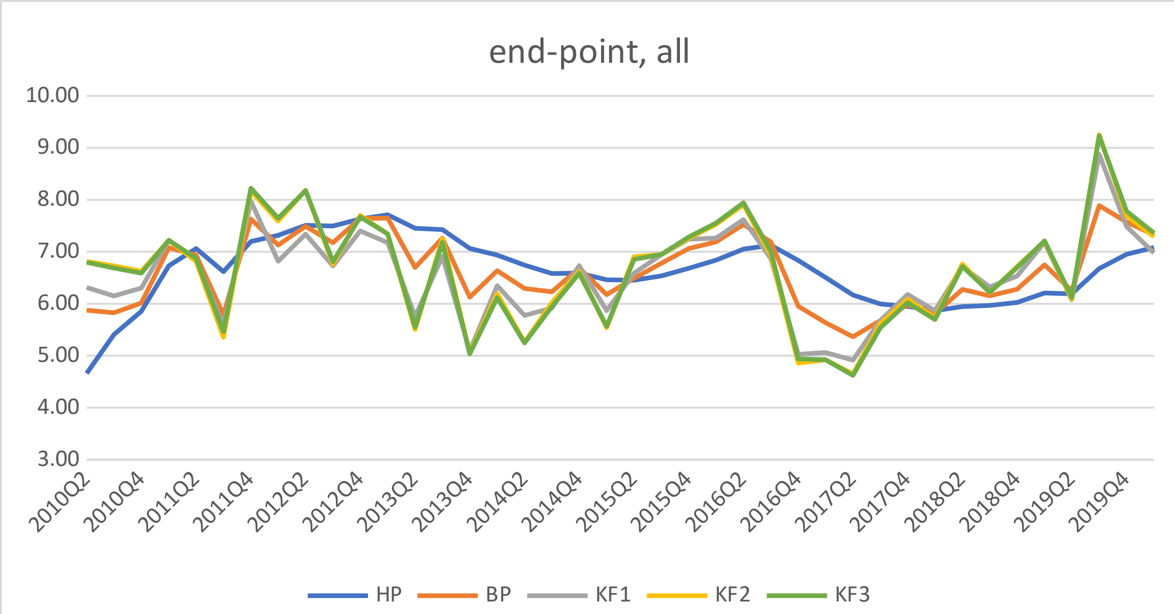
B. Trend output/ long-run growth

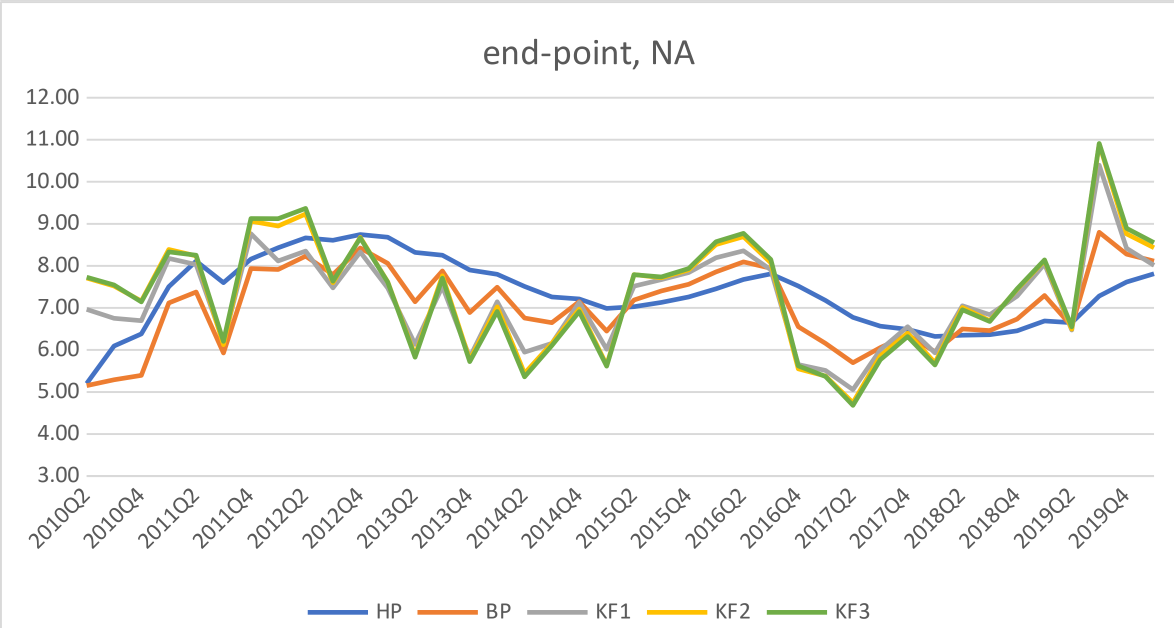
- Figure 2 shows the trend output growth for the different models
  - By design, the smooth results for HP and BP filter are very smooth. However, end-point results are a little bit more volatile.
  - The different KF models give similar results. Except during the start of the sample, where we see a difference between KF1 and KF2+KF3

Figure 2: Trend output growth









- Table 5 shows the average output trend growth over the 2010-2019 time period.
  - End-point results tend to underestimate eventual estimated output trend growth
  - For the smoothed results, KF models have a higher trend growth than the HP and BP models.
  - KF1 has lower average trend output growth than KF2 and KF3
  - Most of these results are driven by the first 8 quarters of the time-period so should be taken with a grain of salt

Table 5: Average trend output growth

data	HP	BP	KF1	KF2	KF3
Smooth, all	6.73	6.69	6.79	6.86	6.87
Endpoint, all	6.64	6.64	6.54	6.63	6.63
Smooth, NA	7.4	7.36	7.52	7.61	7.63
Endpoint, NA	7.35	7.06	7.2	7.33	7.33

- As trend output growth is rather volatile for the Kalman filter, it would be unwise to put too much weight on the estimate of output trend growth in any given quarter. However, using the average over a certain time period might be more informative. Table 6 shows the end-point accuracy of a 5 quarter moving average for output trend growth.
  - The most efficient models differ based on type of output and method used. Most efficient model by data and method is highlighted in green
    - If we use the sum of absolute values, the KF models outperform the HP and BP techniques. With KF3 performing best for all output and KF1 performing best for non-agricultural output.
    - If sum of squares is used, The HP filter is most efficient (although the difference with KF1 performance is very small) and for non-agricultural output KF1 is most efficient.
    - Given the fact that KF1 either is the best performing model or has accuracy very close the best performing model, I think it's safe to say that KF1 is the most efficient overall
  - Results for a 13 quarter moving average (not presented) has KF3 as performing best for all output and KF1 performing best for NA output

Table 6: Difference between smooth and end-point output trend growth (5-quarter moving average)

data	HP	BP	KF1	KF2	KF3
All output					
Sum of absolute values	18.05	16.19	11.66	10.57	10.48
Sum of squares	8.18	14.65	8.24	11.54	12.32
Non-agricultural output					
Sum of absolute values	23.36	27.61	18.65	21.48	22.22
Sum of squares	13.59	17.05	11.2	14.54	14.75

C. Steady-state growth in recent years

- Given the combination of high growth and low inflation, the suggestion was raised that steady-state/ natural output growth might be higher than initially thought. More specially, using average growth levels from the recent past might give a poor estimate of the growth potential.
- Table 7 shows different estimates for natural output growth for 2017-2019 using past output growth, HP, BP and Kalman filters. For the KF models, growth in trend output as well as average estimates for the SSG variable are presented.
  - Important to note is that all figures represent annualised quarterly growth where qoq growth is simply multiplied by 4. (this is done because the benchmark IMF models do it this way). This normally provides a pretty good estimate for proper annual growth but could lead to an underestimation of annual growth if growth is large. Although I doubt this will significantly alter the results, at some point I will have to change this.
  - Using past GDP always gives the lowest estimate for natural output growth
  - Across all methods we see an upward trend in natural output growth
  - The upward trend is steeper for our KF models, increasing the difference between KF models and past GDP+HP+BP based estimates.
    - For the KF models, we see an especially large jump from 2017 to 2018.
    - In 2019, KF2+KF3 tend to provide higher estimates for potential output than KF1

- Given end-point inaccuracy, estimates for 2019 (and potentially 2018) will likely change for the non-GDP based methods as more data comes in. However, given the general superior end-point accuracy of KF1, we can infer that the natural level of output growth has increased recently. Not by as much suggested by the KF2+KF3 models but by more than suggested by past GDP growth averages and the HP and BP models.

Table 7: Estimates for natural output growth

method	All output			Non-agricultural output		
	2017	2018	2019	2017	2018	2019
Past GDP(three-year average)	6.01	6.26	6.62	6.49	6.79	7.23
HP	6.61	6.91	7.07	7.2	7.58	7.79
BP	6.67	6.97	7.2	7.32	7.74	8.07
KF1 (output trend growth)	6.39	7.58	7.64	6.71	8.57	8.79
KF1 (SSG estimate)	6.38	7.42	7.51	6.75	8.37	8.61
KF2 (output trend growth)	6.34	7.65	7.88	6.62	8.67	9.12
KF2 (SSG estimate)	6.34	7.53	7.76	6.67	8.5	8.96
KF3 (output trend growth)	6.27	7.67	7.98	6.49	8.7	9.27
KF3 (SSG estimate)	6.27	7.54	7.85	6.57	8.53	9.11

- In general, a case can be made that steady-state growth has increased recently, which could explain the low inflation and high growth combo in 2019. However, given the weak Philips-curve relationship as measured by our Bayesian estimation, maybe inflation and growth are just not very closely related

D. Forecasting ability of natural growth estimates

- As estimates for natural output growth differ between methods, evaluating the forecasting ability of different methods could provide a good indication of reliability.
    - To properly evaluate forecasting ability, I use average 13 quarter end-point results to forecast average growth for the upcoming 12 quarter. For instance, to evaluate forecasting ability of the HP filter in 2014Q1, I run the filter using data from 2009Q2-2014Q1. I calculate the annualised quarterly growth of the trend then take the average of this growth rate for 2011Q1-2014Q1. Which I then compare to the average realised growth rate for 2014Q2-2017Q1.
    - Unfortunately we only have about 20 observation points and it's a bit crude so maybe different procedures can be explored.

- Table 8 shows forecasting error for the different methods. Both SSG and output trend growth is used to forecast for the KF models.
    - KF1 (SSG) always performs the best
    - Using the SSG variable has better accuracy than using the output trend growth
    - KF2+KF3 (SSG) usually outperform HP and BP but not in all cases.
    - Using past GDP growth normally has the lowest forecasting accuracy even-though end-point accuracy is not an issue here which should already give an edge to this method.

Table 8: Difference between forecasted and realised growth (3-year averages)

data	GDP growth	HP	BP	KF1 (trend)	KF1 (SSG)	KF2 (trend)	KF2 (SSG)	KF3 (trend)	KF3 (SSG)
All output									
Sum of absolute values	16.4	16.1	16.4	13.3	12.9	15.2	14.8	15.3	14.9
Sum of squares	23.0	17.4	21.0	14.7	14.1	17.5	16.8	17.5	16.7
Non-agri output									
Sum of absolute values	19.8	20.0	17.6	15.9	15.4	18.1	17.8	18.3	17.9
Sum of squares	31.4	26.8	26.2	19.8	18.9	24.4	23.4	24.7	23.7

- I have also performed the same exercise using the average natural output growth over 5 quarters to forecast the upcoming 4 quarter (results not shown).
    - Forecasting errors are large across the board
    - The BP and HP filters perform the best, followed closely by KF1(SSG). Past GDP growth has again the lowest accuracy

4. <u>Next steps</u>

Given current results, we have a couple of ways we can proceed with regard to policy briefs or papers. Just to give a few suggestions:

1. Bayesian estimation results and the weak Philips-curve relationship
    - Although interesting maybe not super relevant for MINECOFIN. Probably more something for the Central Bank
    - Probably should include a more sophisticated model and comparison to other countries
2. Superior performance of the Kalman filter compared to the HP and BP filters
    - Would probably not be super well received on its own. So would be best to combine this with suggestion 3 or/and 4. Maybe as an appendix.
3. Output gap and trend growth results with special attention for:
    a. Recent increase in natural output growth estimates
    b. The inferiority of using past GDP growth
4. Once the QPM is complete, maybe a more sophisticated Kalman filter could be developed. It would be interesting to see how much better the more sophisticated model performs compared to our very simple KF1 model.